WEEK 9 Divyadharshini G

Housing Data Cleaning & Transformation

1. Objective

The objective of this task is to clean and transform the King County House Sales dataset using Power BI and Power Query. The project prepares reliable housing data for analysis of property characteristics and sale prices.

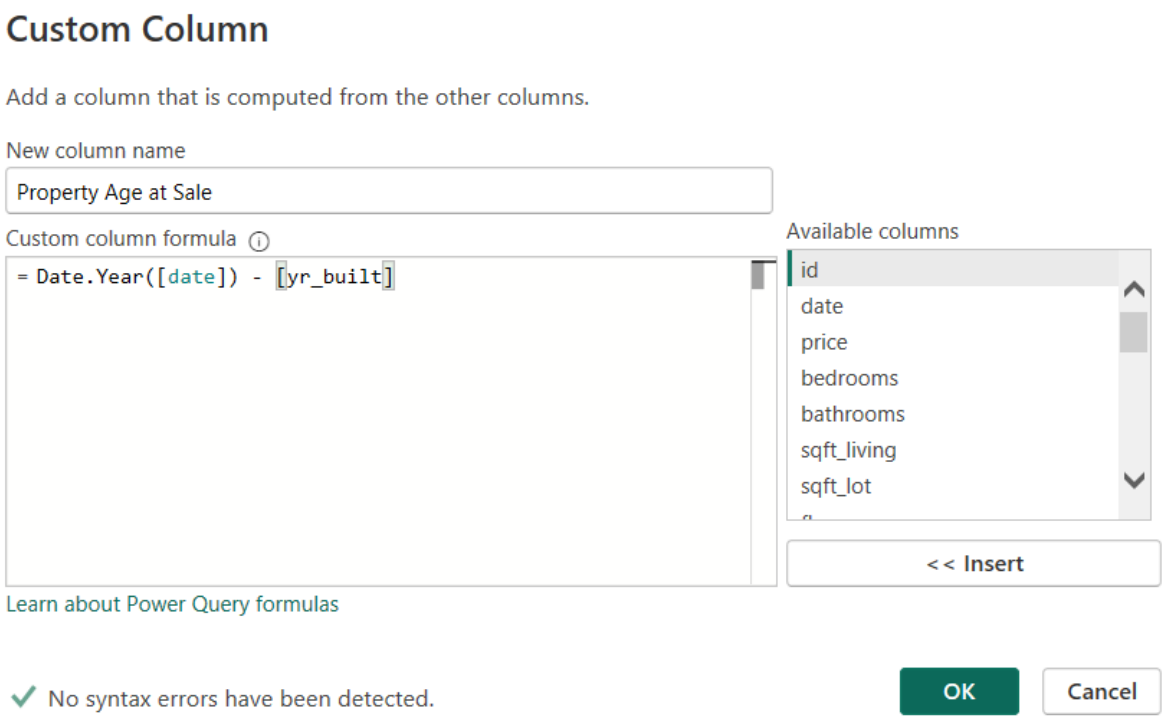
2. Data Cleaning

The dataset was imported into Power Query. Data types were checked, missing values and errors were inspected, duplicate property IDs were checked, and invalid records were filtered. Records with zero bedrooms, zero bathrooms, non-positive prices, and invalid area values were excluded.

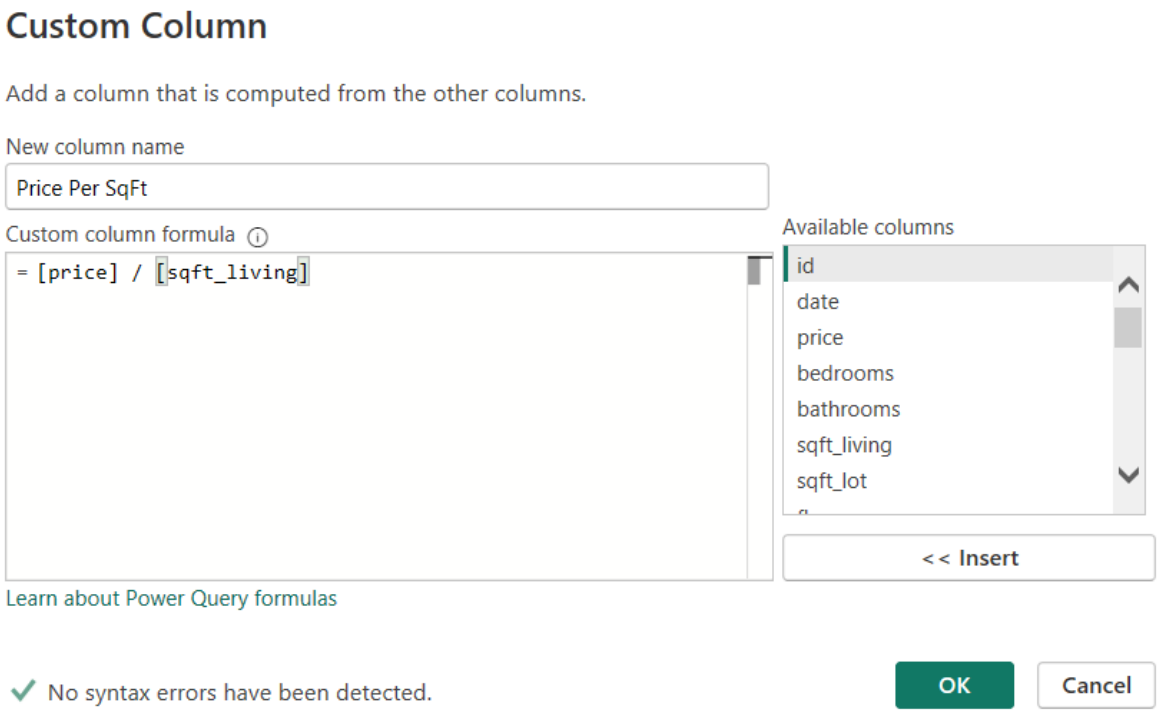
	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	7129300520	13-10-2014 00:00:00	221900	3	1	1180	
2	6414100192	09-12-2014 00:00:00	538000	3	2.25	2570	
3	5631500400	25-02-2015 00:00:00	180000	2	1	770	
4	2487200875	09-12-2014 00:00:00	604000	4	3	1960	
5	1954400510	18-02-2015 00:00:00	510000	3	2	1680	
6	7237550310	12-05-2014 00:00:00	1225000	4	4.5	5420	
7	1321400060	27-06-2014 00:00:00	257500	3	2.25	1715	
8	2008000270	15-01-2015 00:00:00	291850	3	1.5	1060	
9	2414600126	15-04-2015 00:00:00	229500	3	1	1780	
10	3793500160	12-03-2015 00:00:00	323000	3	2.5	1890	
11	1736800520	03-04-2015 00:00:00	662500	3	2.5	3560	
12	9212900260	27-05-2014 00:00:00	468000	2	1	1160	
13	114101516	28-05-2014 00:00:00	310000	3	1	1430	
14	6054650070	07-10-2014 00:00:00	400000	3	1.75	1370	
15	1175000570	12-03-2015 00:00:00	530000	5	2	1810	
16	9297300055	24-01-2015 00:00:00	650000	4	3	2950	
17	1875500060	31-07-2014 00:00:00	395000	3	2	1890	
18	6865200140	29-05-2014 00:00:00	485000	4	1	1600	
19	16000397	05-12-2014 00:00:00	189000	2	1	1200	
20	7983200060	24-04-2015 00:00:00	230000	3	1	1250	
21	6300500875	14-05-2014 00:00:00	385000	4	1.75	1620	
22	2524049179	26-08-2014 00:00:00	2000000	3	2.75	3050	
23	7137970340	03-07-2014 00:00:00	285000	5	2.5	2270	
24	8091400200	16-05-2014 00:00:00	252700	2	1.5	1070	

3. Calculated Columns

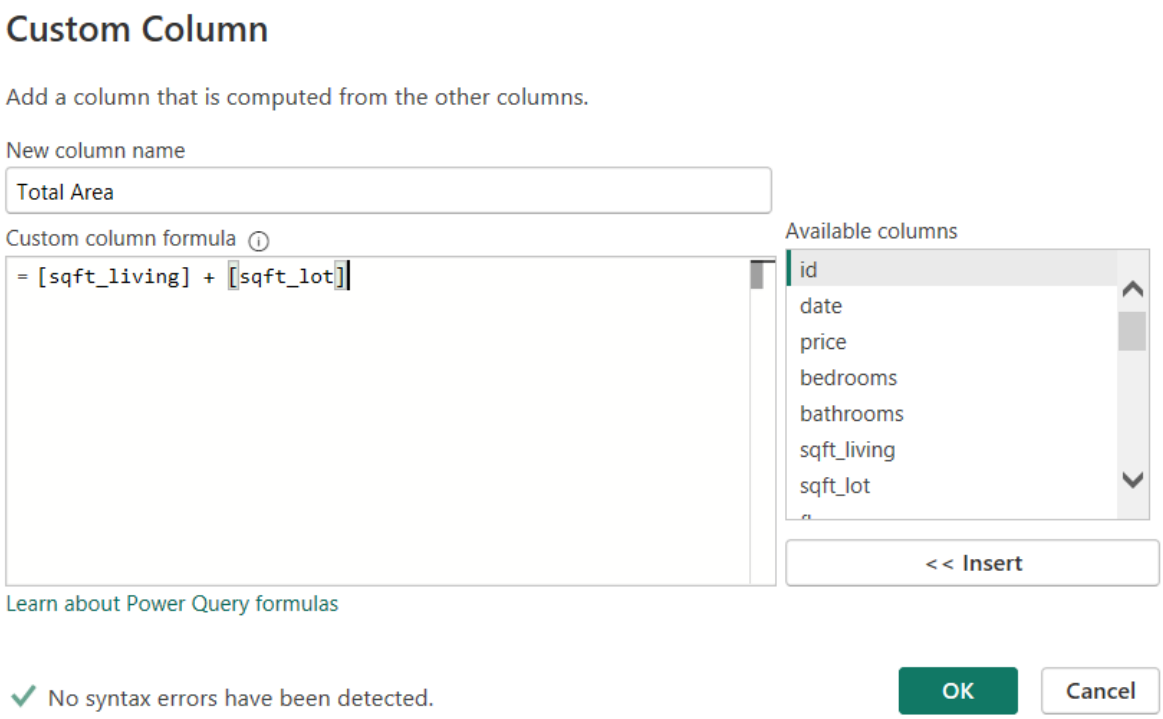
1. Property Age at Sale: Calculates the age of a house at the time of sale using the sale year and construction year.



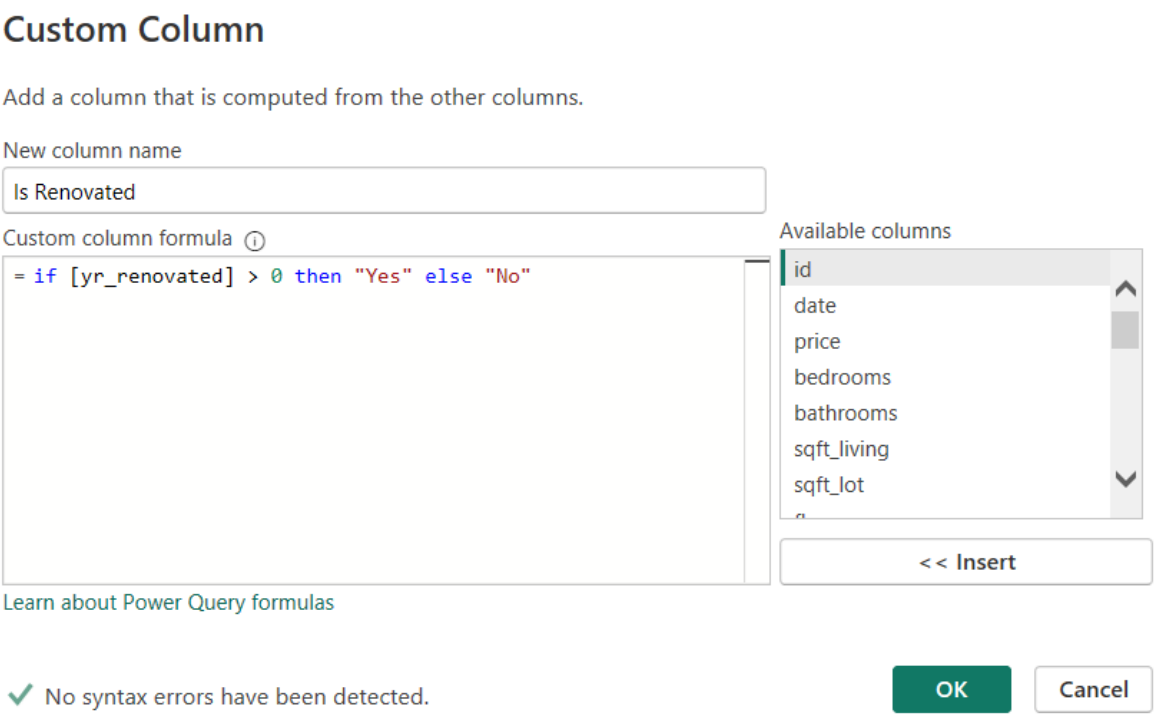
2. Price per SqFt: Calculates the price per square foot by dividing the house price by its living area.



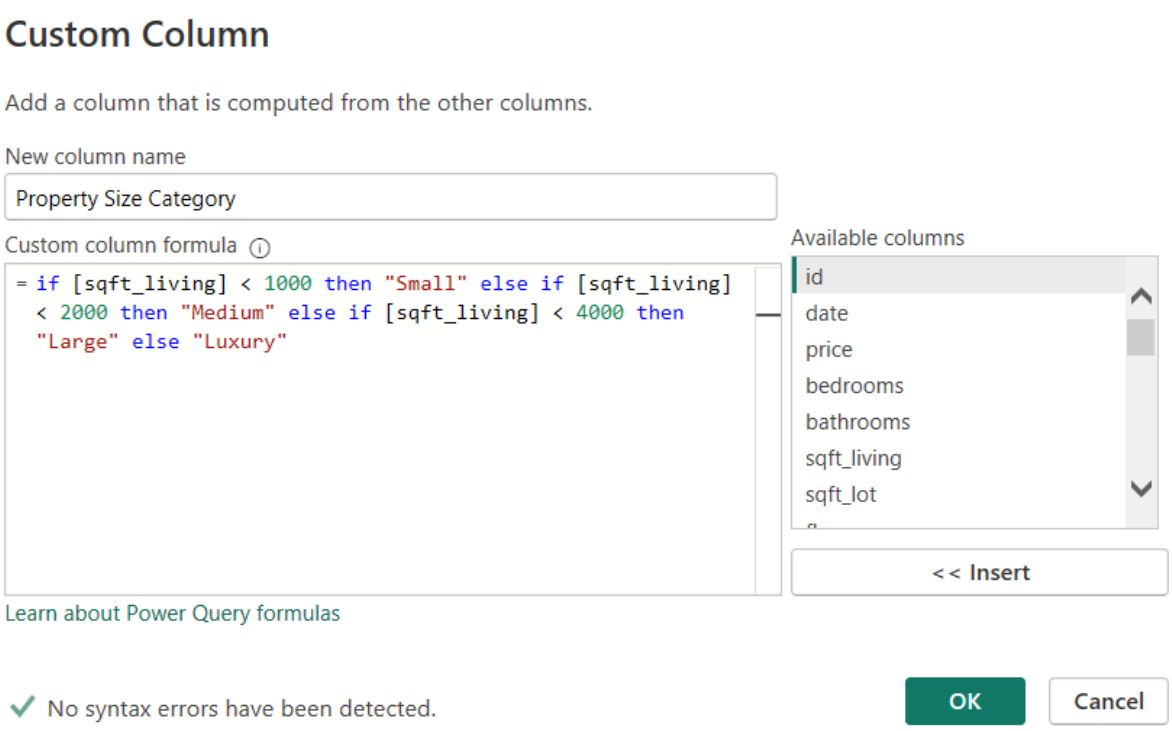
3. Total Area: Adds the living area and lot area to create a combined area value for this project.



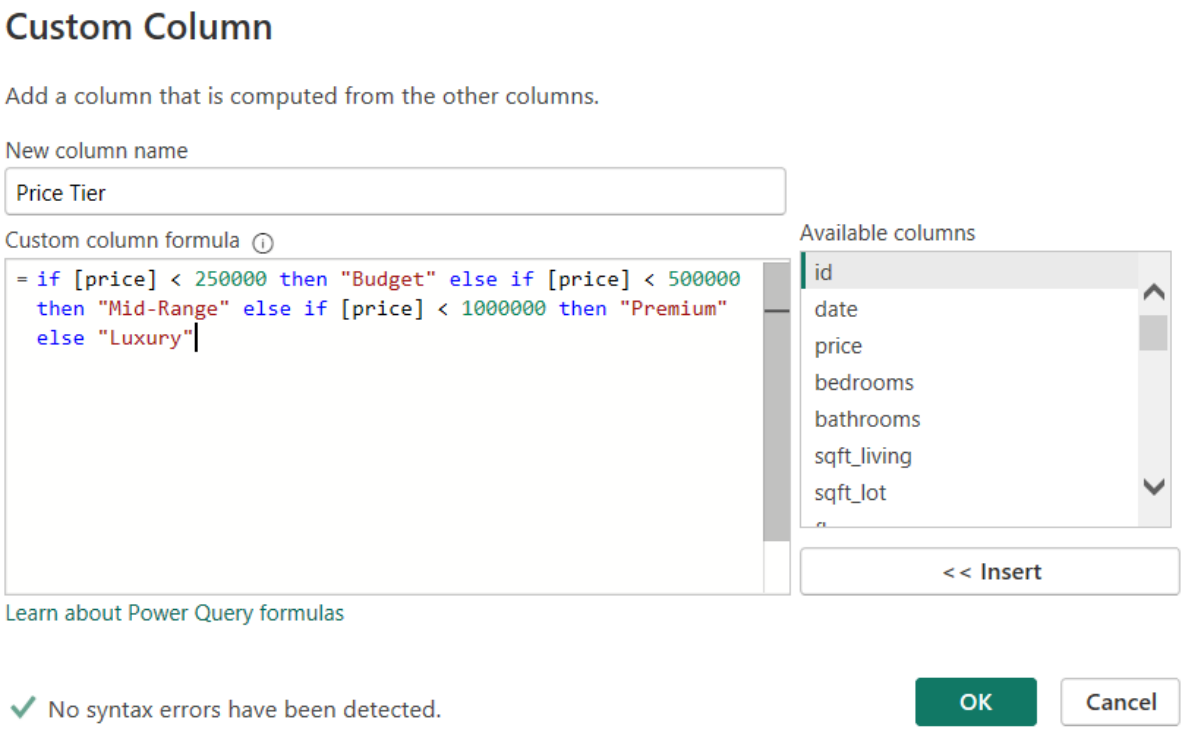
4. Is Renovated: Classifies houses as renovated or non-renovated based on whether a renovation year is recorded.



5. Property Size Category: Groups houses into Small, Medium, Large, and Luxury categories based on living area.



6. Price Tier: Groups houses into Budget, Mid-Range, Premium, and Luxury categories based on their prices.



4. Dashboard Visualizations

Average House Price: Displays the average price of houses.

Total Houses: Displays the number of housing records included after filtering.

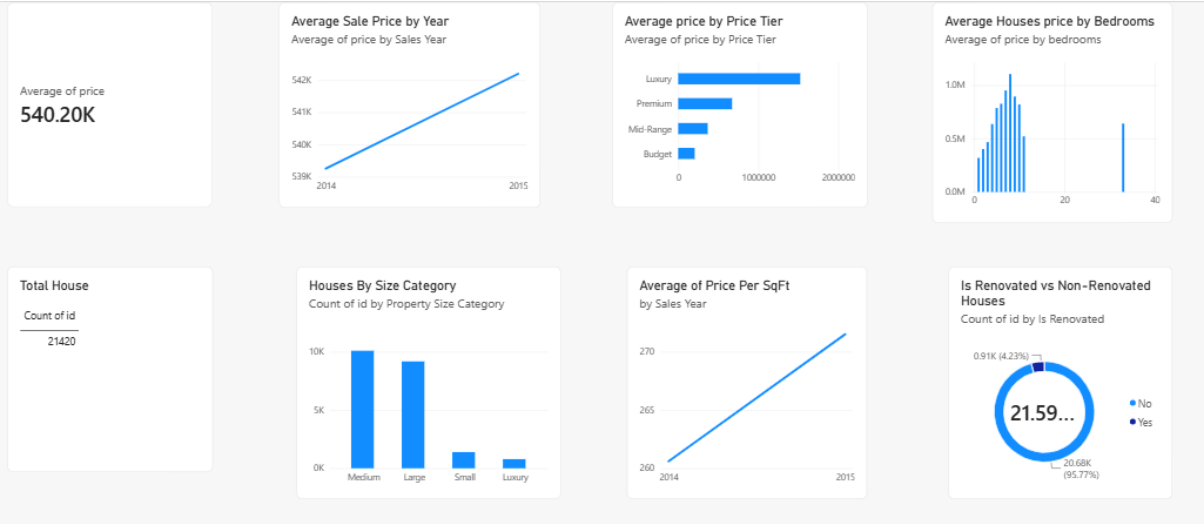
Houses by Size Category: Compares the number of houses in each property size category.

Average Sale Price by Year: Shows the change in average house prices over time.

Renovated vs Non-Renovated Houses: Compares the number of renovated and non-renovated properties.

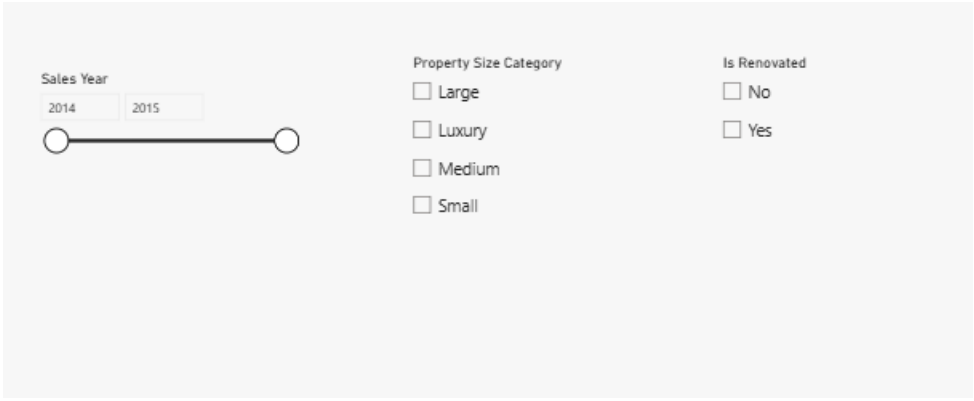
Average House Price by Bedrooms: Shows the relationship between bedroom count and average house price.

Average Price per SqFt by Year: Displays changes in average price per square foot over time.



5. Slicers

- **Sales Year:** Filters the data by sale year.
- **Is Renovated:** Filters houses based on renovation status.
- **Property Size Category:** Filters houses by size category.



6. Final Dashboard

